

Experiment No. 21

Title

Prolog Program to Implement Towers of Hanoi

Aim

To write a Prolog program to implement the **Towers of Hanoi** problem.

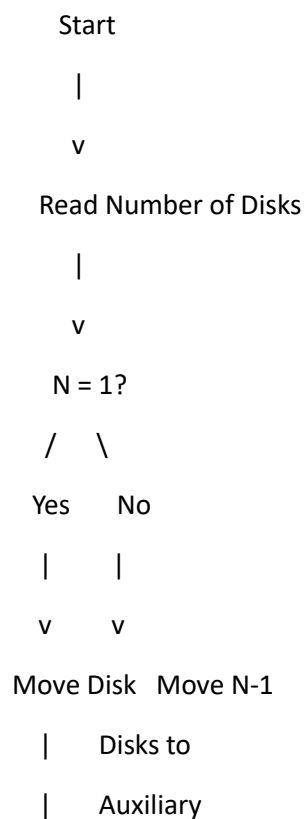
Objective

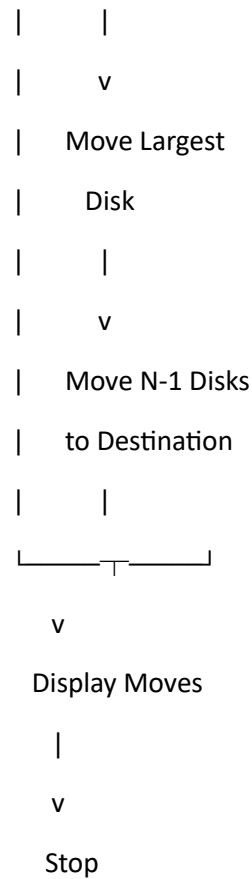
The objective of this experiment is to understand the use of **recursion in Prolog** to solve the Towers of Hanoi problem.

Algorithm

1. Start.
2. If there is one disk, move it directly from source to destination.
3. Move N-1 disks from source to auxiliary.
4. Move the remaining disk from source to destination.
5. Move N-1 disks from auxiliary to destination.
6. Display the moves.
7. Stop.

Flowchart





Prolog Program

hanoi(1, Source, Destination, _) :-

```

write('Move disk from '),
write(Source),
write(' to '),
write(Destination),
nl.

```

hanoi(N, Source, Destination, Auxiliary) :-

```

N > 1,
N1 is N - 1,
hanoi(N1, Source, Auxiliary, Destination),
write('Move disk from '),
write(Source),
write(' to '),
write(Destination),

```

nl,

hanoi(N1, Auxiliary, Destination, Source).

Query

?- hanoi(3,left,right,middle).

Sample Output

Move disk from left to right

Move disk from left to middle

Move disk from right to middle

Move disk from left to right

Move disk from middle to left

Move disk from middle to right

Move disk from left to right

true.

Result

The Prolog program successfully implemented the **Towers of Hanoi** problem using recursion.

Conclusion

Thus, the Towers of Hanoi problem was successfully solved in Prolog by recursively moving the disks between the three rods.